A Shakti Rao

Summary

MCA student (2027 batch) building full-stack JavaScript applications with Node.js, Express, React and MongoDB. Built a user-scoped REST API with JWT cookie authentication, email OTP verification and Zod validation, and deployed a music app with media uploads and an admin dashboard. Comfortable debugging real production-style issues including diagnosing and fixing a token-refresh retry loop and eager to grow into an engineering role.

Skills

- **Languages**: Javascript(ES6+), Typescript, Core Java
- **Frontend**: React.js, Vite, Tailwind CSS, TanStack React Query, shadcn/ui, Bootstrap, Responsive UI design, HTML, CSS
- **Backend**: Node.js, Express.js, RESTful API Design, JWT Authentication, bcrypt, Zod (schema validation), Multer file uploads,Nodemailer, CORS, rate limiting
- **Database**: SQL, Postgres, MongoDB, Mongoose, Schema Design, compound / TTL indexes CRUD Operations, Aggregation Pipeline, Schema Design
- **Tools & Practices**: Git, GitHub, Docker, Docker Compose, Postman, Cloudinary, VS Code, Vercel, Agile collaboration, Debugging, npm
- **Coursework**: Data Structures & Algorithms, Operating Systems, Database Management, Networking, OOP

Projects

Expense Tracker — Full-Stack Personal Finance Application *(In Progress)*
React.js · Node.js · Express.js · MongoDB · Docker · Tailwind CSS — GitHub

- Built 20+ REST endpoints for users, categories and expenses, with every query scoped to the logged-in user and shared Zod validation middleware for request bodies and query parameters.
- Implemented JWT access and refresh authentication with bcrypt password hashing and httpOnly, secure, SameSite=Strict cookies. Added email OTP verification and password reset through Nodemailer, with hashed OTPs that expire automatically through a MongoDB TTL index.
- Added server-side pagination, sorting and filtering (category, date range, amount range, payment method, title search) to the expenses API, supported by compound indexes on user and date and on user and category.
- Fixed an infinite retry loop in the token-refresh flow by separating the Axios instance that performs the refresh from the one carrying the response interceptor.
- Built the React frontend with React Query for data fetching, caching and invalidation, plus custom categories with icon uploads, bulk delete and category reassignment. Containerized the backend, frontend and MongoDB with Docker Compose.

Musicify — Full-Stack Music Streaming Application *(Live)*
React.js · Vite · Node.js · Express.js · MongoDB · Tailwind CSS · Cloudinary — Live Demo | GitHub

- Built the app as three parts: a React player, a React admin dashboard and an Express and MongoDB API, deployed on Vercel.
- Implemented song and album upload endpoints with Multer (MIME-type filtering, 50 MB limit) and Cloudinary storage for audio and cover images, with temporary-file cleanup after every request.
- Secured the API with Helmet, a CORS origin allowlist and rate limiting (100 requests per 15 minutes), and added centralized handling for upload and server errors.
- Built the player controls (play, pause, next, previous, seek bar) with React Context, and an admin interface to add and remove songs and albums.

Education

- **Lovely Professional University** — Phagwara, India
  Master's of Computer Application - Full Stack Development — *Jan 2025 - Jan 2027*
  Courses: *Operating Systems, Data Structures, Analysis Of Algorithms, Full Stack Development, Database Management, Networking*
- **University of Calcutta** — Kolkata, India
  Bachelor's of Science - Computer Application — *July 2021 - July 2024*
  Courses: *Java, Object-Oriented Programming,*

Academic Competencies & Certifications

- **DSA & OOP**: Arrays, strings, recursion, searching & sorting | Java & JavaScript – classes, inheritance, abstraction, modular design
- **Web & Database**: Frontend–backend integration, RESTful APIs, client–server architecture | MongoDB, CRUD operations, relational concepts, data modeling
- **Systems & Engineering**: Process management, server-side execution, memory concepts | SDLC, version control, debugging, testing, requirement analysis
- **Certifications Achievements**: ReactJS for Beginners (Dec 2024) · JavaScript Intermediate, HackerRank (Feb 2025) · GenAI Essentials (Oct 2026) · LeetCode: 107 problems solved across arrays, strings, recursion, binary search and linked lists